

AARADHYA DEV TAMRAKAR

AI/ML · Edge AI · Intelligent Systems · Robotics

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PROFILE

Electronics, Communication and Information Engineering student at IOE, Kathmandu Engineering College, focused on AI/ML, edge AI, responsible AI, and intelligent systems. Builds end-to-end machine learning systems spanning model development, statistical evaluation, explainability, APIs, and deployment, with hands-on experience in ESP32-S3, computer vision, embedded control, and robotics. Currently developing BiasAperture, a fairness-auditing framework for facial-analysis systems, and SPARK, an ESP32-S3 edge-AI wearable fall-detection system.

EDUCATION

Bachelor of Engineering (B.E.)

Expected: January 2027

Electronics, Communication and Information — Institute of Engineering (IOE), Kathmandu Engineering College | Year IV / Part II

Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems

Cumulative Percentage: 75%

EXPERIENCE

Vice Chair

2026 – Present

IEEE KEC KTM Student Branch

- Directed branch strategy, scaling initiatives and member engagement across executive committees.
- Managed relations across the broader technical community.

Event Manager

2026 – Present

Electronics Project Club (EPC), Kathmandu Engineering College

- Coordinated logistics for club workshops and department activities.

Makerspace Ambassador

June 2026 – Present

KEC Maker's Space, Kathmandu Engineering College

- Shipped site-wide fixes to the Maker's Space website — image repair, content consistency, UI cleanup.

Vice Secretary

2025 – 2026

IEEE KEC KTM Student Branch

- Streamlined operations and grew program participation by managing documentation and orchestrating workshop logistics.
- Coordinated volunteers alongside the committee for seamless execution.

Resource Manager

2024 – 2026

Electronics Project Club (EPC), Kathmandu Engineering College

- Allocated scheduling and resources for workshops across multiple departments.

FELLOWSHIPS & RECOGNITIONS

Fuse AI Fellow — Fusemachines AI Fellowship 2026

Active

Fusemachines

- Engineered a five-stage agentic Text-to-SQL pipeline (Planner → Generator → Validator → Executor → Summarizer), achieving 100% execution success and accuracy on a 50-question benchmark.
- Built containerized AI services using FastAPI, PostgreSQL, and Docker.
- Developed a churn/CLV pipeline achieving 5-fold CV ROC-AUC 0.841, with threshold tuning and SHAP explainability.
- Applied Random Forest, XGBoost, Bayesian inference, Gaussian processes, clustering, and forecasting across applied datasets.
- Worked with UCI Online Retail II (~500K transactions) for segmentation and forecasting, achieving MASE 2.44 with a four-model ensemble.

NSSR DataCamp Fellow — Cohort 2

2026 – Present

Nepalese Society of Student Researchers

- Secured competitive selection (April 2026) for a sponsored DataCamp Premium license, one of a limited cohort.
- Delivered six applied GenAI solutions, automating null filtering, deduplication, and schema normalization across product and data workflows.
- Completed coursework toward SQL Associate and Python Data Associate certifications in PostgreSQL, Pandas, statistics, and scikit-learn.

PROJECTS

BiasAperture — Vision Fairness & Bias Audit (Fellowship Capstone)

AI360 · Fairlearn · FairFace · UTKFace · SHAP · Statistical Testing

- Developed a fairness-auditing framework for facial-analysis classifiers, generating structured reports aligned with EU AI Act Article 10 and NIST AI RMF concepts.
- Computes DPD, EOD, EOP, and DIR across 126 demographic bins (race $\times$ age $\times$ gender) with χ^2 significance tests and BCa bootstrap confidence intervals.
- Fusemachines AI Fellowship Capstone, two-person team with Tisha Manandhar; 4 of 5 milestones complete (Schema Lock through Statistical Engine).

SPARK — Wearable Fall Detection System (Major Project)

ESP32-S3 · TensorFlow Lite Micro · SHAP · MQTT · FastAPI · PostgreSQL

- Designed a two-layer ESP32-S3 fall-detection system combining a <5ms threshold gate with an INT8 1D CNN confirmation layer under 100ms, reaching 0.919 AUC-ROC on the SisFall benchmark (subject-grouped 5-fold CV, 38,420 windows).
- Computes per-event SHAP explanations at a local, non-cloud gateway; proposal presented to the review panel, demo targeted for March 2027.
- Led the AI and backend work package (WP 1.0) — CNN training pipeline, FastAPI/PostgreSQL gateway — within a 4-person team supervised by Er. Dipen Manandhar.

Gesture-Controlled Self-Balancing Robot (GCSBR) — Minor Project

Computer Vision · Arduino · MPU6050 · Stepper Motors · Android · MATLAB

- Implemented PID stabilization firmware, validated via a MATLAB-simulation-to-hardware pipeline.
- Developed an Android gesture-control app (MediaPipe, CameraX, HC-05 BT) driving motors for real-time actuation.

Edge AI Stability Detection System

Python · Scikit-learn · FastAPI · Joblib

- Predicted platform stability from simulated IMU data using a Random Forest model, reaching 99.8% test accuracy; served via FastAPI with Joblib export for integration into GCSBR.

Alpha Android Super-App

Kotlin · Jetpack Compose · Material3 · DataStore · Apache POI

- Modular Android super-app (Kotlin/Jetpack Compose, Material3) — gesture remote, budget tracker, and multi-mode calculator; calculator is the primary shipping target, with Play Store release in progress.

Nexus — Personal AI Operating System

React (Vite) · FastAPI · SQLite + FTS5 · httpx · asyncio

- Unified Groq (Llama 3.3 70B) and Gemini into an AI workspace via parallel fan-out routing, with SQLite + FTS5 context injection.

Super-NLM — Multi-Account NotebookLM Hub & MCP Server

Python · MCP · NotebookLM · FastAPI · Round-Robin Rate-Limit Routing

- Built an MCP-enabled dashboard for parallel NotebookLM access across N Google accounts, with round-robin account routing and quota cooldown.

SKILLS

Programming: Python, C, C++, Kotlin, SQL, VHDL

AI/ML: scikit-learn, XGBoost, Random Forest, SHAP, AIF360, Fairlearn, PyMC, ArviZ, pgmpy, clustering, forecasting

AI Systems: FastAPI, Docker, PostgreSQL, SQLite/FTS5, REST APIs, asyncio, MCP

Computer Vision/Mobile: OpenCV, MediaPipe, CameraX, Kotlin, Jetpack Compose

Embedded/Robotics: ESP32-S3, Arduino, MPU6050, Stepper Motors, UART, MQTT, FPGA/Vivado

Tools: Git, GitHub, MATLAB, Jupyter, Colab, LaTeX/Overleaf

CERTIFICATIONS & ACTIVITIES

IEEE WIE Nepal LaTeX Workshop (May 2026) · IEEE SPAX organizing team · Prompt Engineering Fundamentals, TechAxis · IEEEExtreme 19.0 (Team ShadowXTREME) · PCB Design & Fabrication Workshop, KEC Robotics Club · AWS Cloud Computing Workshop · Microsoft Learn Student Ambassador

Active in IEEE KEC, EPC, KEC Maker's Space, and KEC Music Club. | Last Updated: August 24, 2026